Probability, Distributions, and Loss Functions Formula Sheet

Probability Rules

$$0 \leq P(A) \leq 1, \quad P(\Omega) = 1$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Example: Bag with 4 red and 6 blue balls, draw 2 without replacement:

$$P(2 \text{ red}) = \frac{4}{10} \cdot \frac{3}{9} = \frac{2}{15}$$

Expectation and Variance

Discrete:

$$E[X] = \sum_x x \cdot p(x)$$

Continuous:

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

Variance:

$$Var(X) = E[X^2] - (E[X])^2$$

Example (fair die):

$$E[X] = 3.5, \quad Var(X) = \frac{35}{12}$$

Common Distributions

Bernoulli

$$E[X] = p, \quad Var(X) = p(1-p)$$

Binomial

$$E[X] = np, \quad Var(X) = np(1-p)$$

Poisson

$$E[X] = \lambda, \quad Var(X) = \lambda$$

Normal

$$E[X] = \mu, \quad Var(X) = \sigma^2$$

Regression Loss Functions

MSE:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

MAE:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Huber:

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y-\hat{y})^2 & |y-\hat{y}| \leq \delta \\ \delta|y-\hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases}$$

RSS:

$$\text{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Classification Loss Functions

Cross-Entropy:

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$$

Hinge:

$$L = \sum_{i=1}^{n}\max(0, 1 - y_i\hat{y}_i)$$

KL Divergence:

$$D_{KL}(P||Q) = \sum_i P(i)\log\frac{P(i)}{Q(i)}$$

Summary Comparison Table

Concept	**Expectation**	**Variance**
Bernoulli(p)	p	$p(1-p)$
Binomial(n, p)	np	$np(1-p)$
Poisson(λ)	λ	λ
Normal(μ, σ^2)	μ	σ^2
MSE	Avg. squared error	Sensitive to outliers
MAE	Avg. absolute error	Robust to outliers
Huber	Hybrid quadratic/linear	Balances sensitivity
Cross-Entropy	Prob. classification error	Penalizes wrong confidence